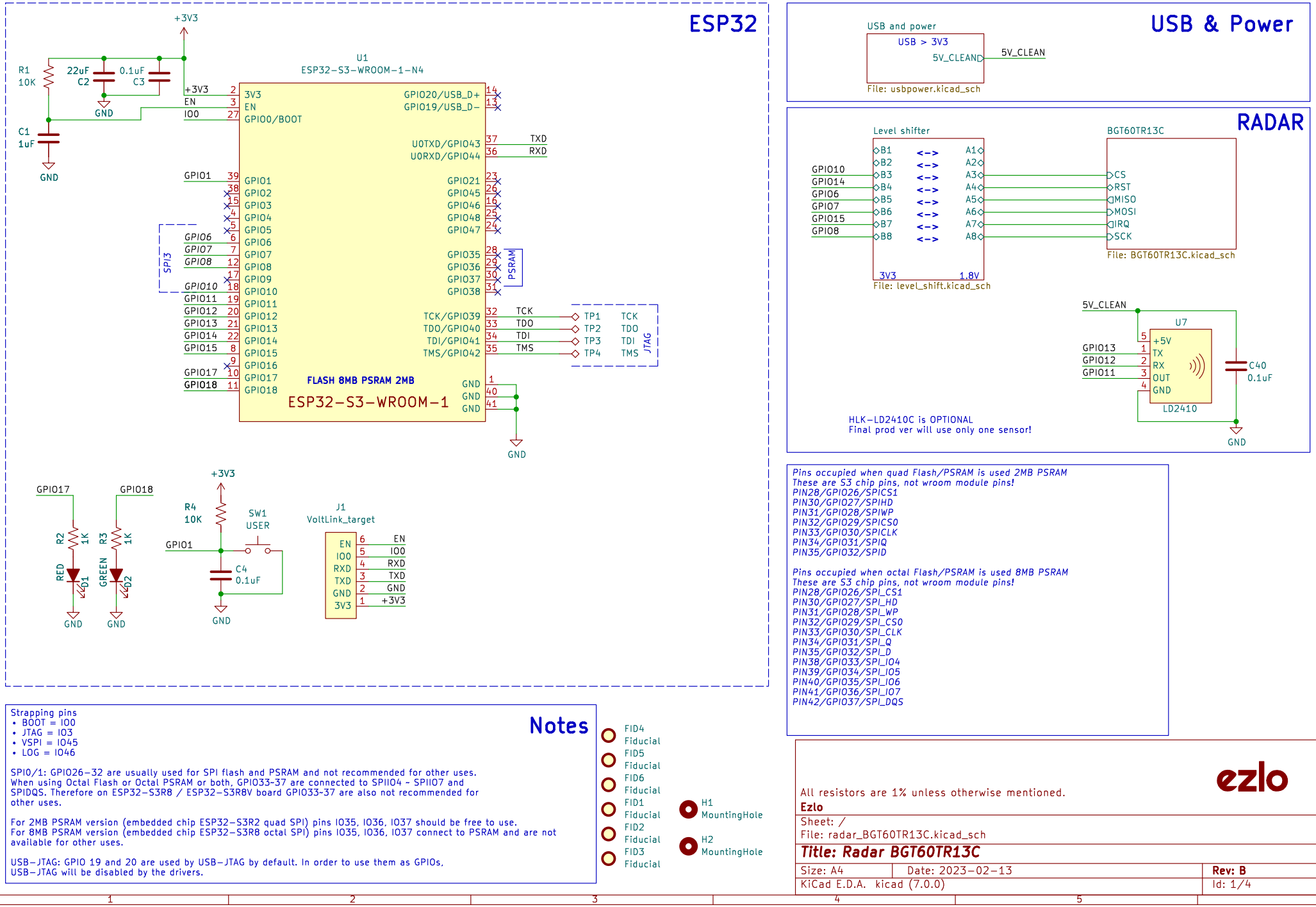
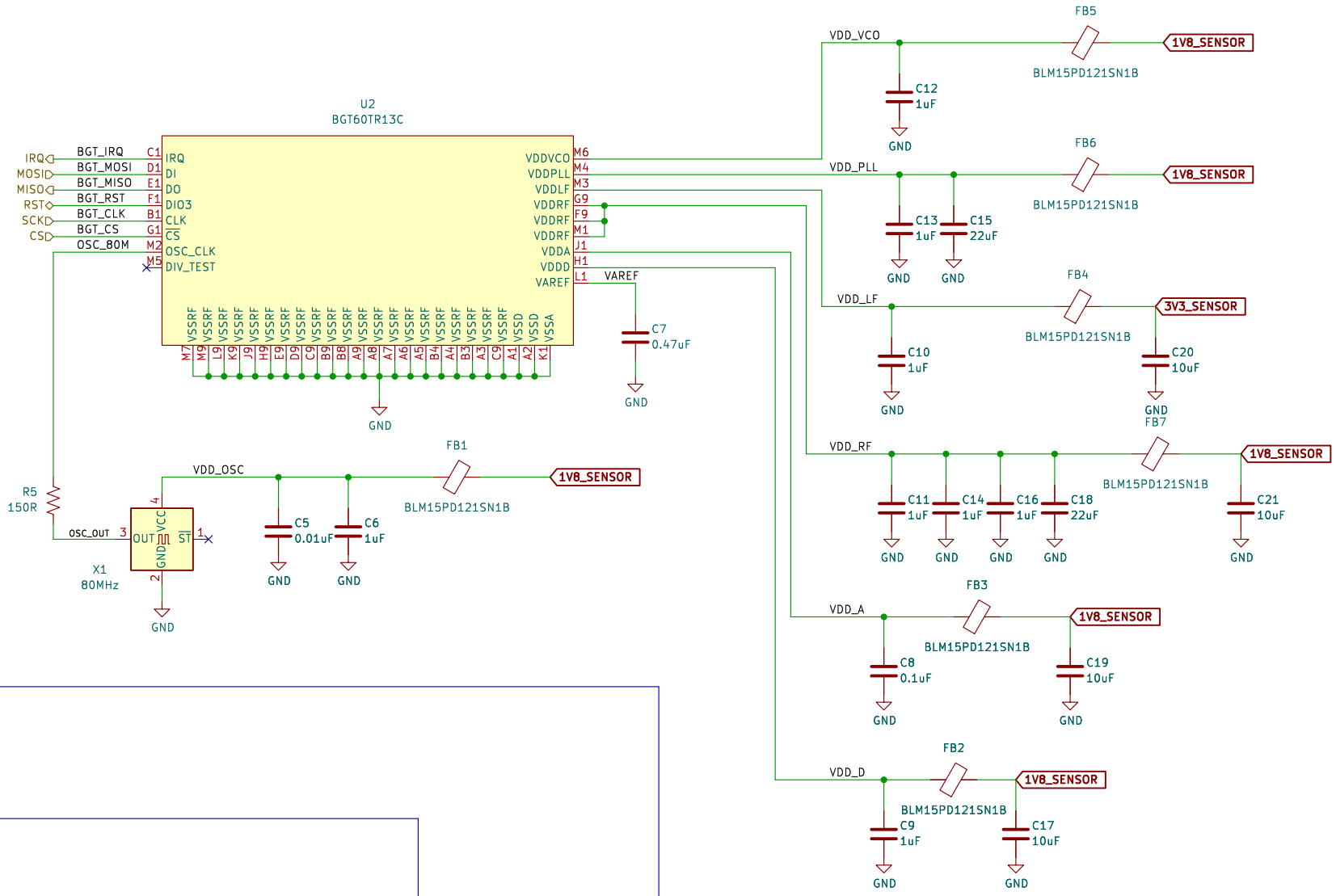

ESP32
U1
ESP32-S3-WROOM-1-N4
FLASH 8MB PSRAM 2MB
ESP32-S3-WROOM-1
+3V3
R1
10K
22uF
C2
0.1uF
C3
C1
1uF
GND
EN
IO0
GPIO0/BOOT
3V3
GPIO1
GPIO2
GPIO3
GPIO4
GPIO5
GPIO6
GPIO7
GPIO8
GPIO9
GPIO10
GPIO11
GPIO12
GPIO13
GPIO14
GPIO15
GPIO16
GPIO17
GPIO18
GPIO19/USB_D-
GPIO20/USB_D+
U0RXD/GPIO44
U0TXD/GPIO43
RXD
TXD
GPIO21
GPIO45
GPIO46
GPIO48
GPIO47
GPIO35
GPIO36
GPIO37
GPIO38
PSRAM
SPI3
TCK/GPIO39
TDO/GPIO40
TDI/GPIO41
TMS/GPIO42
TCK
TDO
TDI
TMS
JTAG
TP1
TP2
TP3
TP4
GND
R2
1K
R3
1K
RED
GREEN
D1
D2
R4
10K
SW1
USER
C4
0.1uF
J1
VoltLink_target
USB & Power
USB and power
USB > 3V3
5V_CLEAN
File: usbpower.kicad_sch
RADAR
Level shifter
BGT60TR13C
B1
B2
B3
B4
B5
B6
B7
B8
A1
A2
A3
A4
A5
A6
A7
A8
CS
RST
MISO
MOSI
IRQ
SCK
3V3
1.8V
File: level_shift.kicad_sch
File: BGT60TR13C.kicad_sch
U7
+5V
TX
RX
OUT
GND
LD2410
C40
0.1uF
HLK-LD2410C is OPTIONAL
Final prod ver will use only one sensor!
Pins occupied when quad Flash/PSRAM is used 2MB PSRAM
These are S3 chip pins, not wroom module pins!
PIN28/GPIO26/SPICS1
PIN30/GPIO27/SPIHD
PIN31/GPIO28/SPIWP
PIN32/GPIO29/SPICS0
PIN33/GPIO30/SPICLK
PIN34/GPIO31/SPIQ
PIN35/GPIO32/SPID
Pins occupied when octal Flash/PSRAM is used 8MB PSRAM
These are S3 chip pins, not wroom module pins!
PIN28/GPIO26/SPI_CS1
PIN30/GPIO27/SPI_HD
PIN31/GPIO28/SPI_WP
PIN32/GPIO29/SPI_CS0
PIN33/GPIO30/SPI_CLK
PIN34/GPIO31/SPI_Q
PIN35/GPIO32/SPI_D
PIN38/GPIO33/SPI_IO4
PIN39/GPIO34/SPI_IO5
PIN40/GPIO35/SPI_IO6
PIN41/GPIO36/SPI_IO7
PIN42/GPIO37/SPI_DQS
Notes
Strapping pins
• BOOT = IO0
• JTAG = IO3
• VSPI = IO45
• LOG = IO46
SPI0/1: GPIO26-32 are usually used for SPI flash and PSRAM and not recommended for other uses.
When using Octal Flash or Octal PSRAM or both, GPIO33~37 are connected to SPIIO4 ~ SPIIO7 and SPIDQS. Therefore on ESP32-S3R8 / ESP32-S3R8V board GPIO33~37 are also not recommended for other uses.
For 2MB PSRAM version (embedded chip ESP32-S3R2 quad SPI) pins IO35, IO36, IO37 should be free to use.
For 8MB PSRAM version (embedded chip ESP32-S3R8 octal SPI) pins IO35, IO36, IO37 connect to PSRAM and are not available for other uses.
USB-JTAG: GPIO 19 and 20 are used by USB-JTAG by default. In order to use them as GPIOs, USB-JTAG will be disabled by the drivers.
FID4 Fiducial
FID5 Fiducial
FID6 Fiducial
FID1 Fiducial
FID2 Fiducial
FID3 Fiducial
H1 MountingHole
H2 MountingHole
All resistors are 1% unless otherwise mentioned.
Ezlo
Sheet: /
File: radar_BGT60TR13C.kicad_sch
Title: Radar BGT60TR13C
Size: A4
Date: 2023-02-13
Rev: B
KiCad E.D.A. kicad (7.0.0)
Id: 1/4

RADAR SENSOR





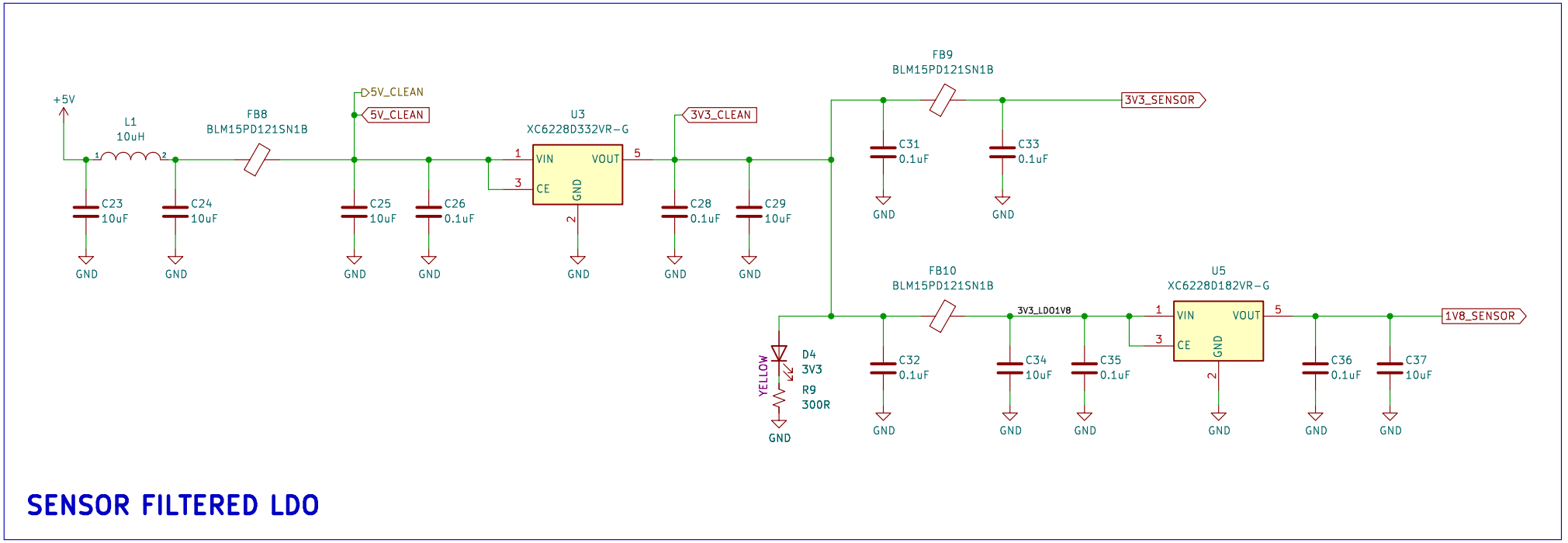
All resistors are 1% unless otherwise mentioned.

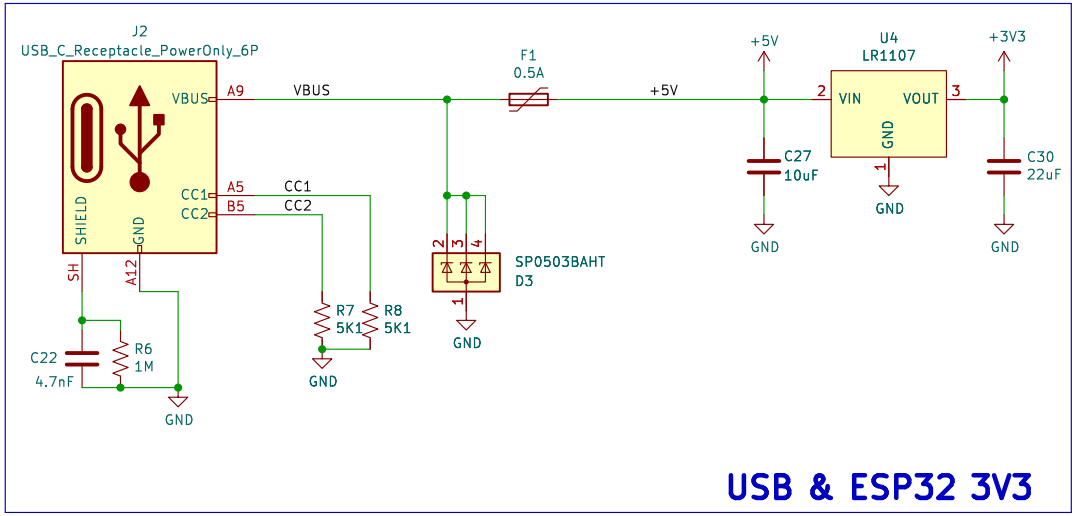
Ezio

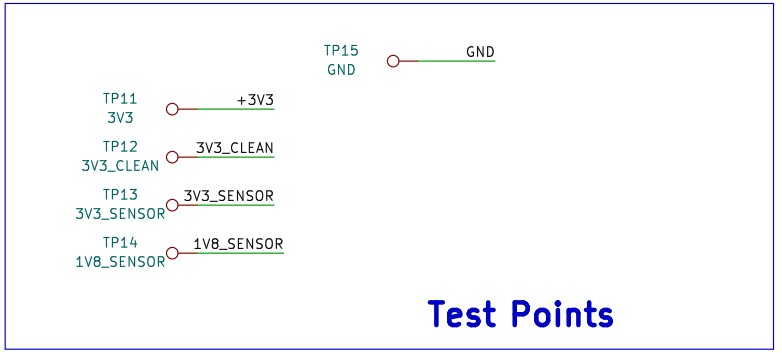
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File: BGT60TR13C.kicad_sch

Title: Radar BGT60TR13C

Size: A4	Date: 2023-02-13	**Rev: B**
KiCad E.D.A. kicad (7.0.0)		Id: 3/4







All resistors are 1% unless otherwise mentioned.

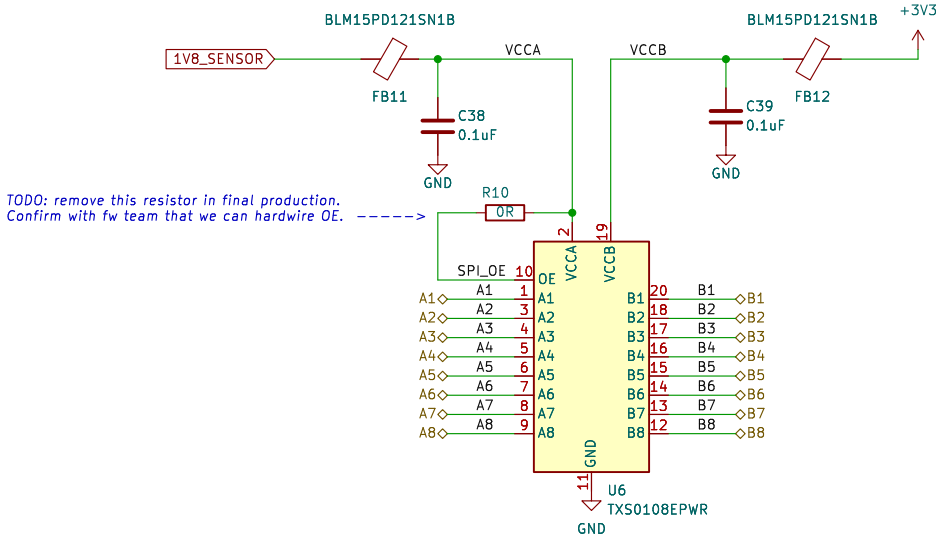
Ezio

Sheet: /USB and power/
File: usbpower.kicad_sch

Title: Radar BGT60TR13C

Size: A4	Date: 2023-02-13	**Rev: B**
KiCad E.D.A. kicad (7.0.0)		Id: 4/4

Level Shifter



1.4 V to 3.6 V on A port and 1.65 V to 5.5 V on B port (VCCA ≤ VCCB)!

All resistors are 1% unless otherwise mentioned.

Ezlo

Sheet: /Level shifter/
File: level_shift.kicad_sch

Title: Radar BGT60TR13C

Size: A4	Date: 2023-02-13	**Rev: B**
KiCad E.D.A. kicad (7.0.0)		Id: 5/4